CORE PRODUCT

PROBLEMS

Cycling experiences are difficult to translate into measurable evidence
Cyclists experience unsafe overtaking, limited cycling space, poor surfaces and environmental conditions, but these lived experiences are difficult to document systematically and translate into evidence for collective discussion.

SOLUTIONS

From lived experience to measurable data
The senseBox:bike protocol connects cyclists' lived experiences with georeferenced sensor measurements, helping document conditions such as lateral distance, available cycling space, road roughness and environmental conditions.

PROBLEMS

Cycling safety is not only an infrastructure issue
Cycling conditions depend not only on bike lanes, but also on interactions between cyclists, drivers, pedestrians, street design, parking and traffic behaviour.

SOLUTIONS

From data to shared interpretation
Transform raw data into collective insights through workshops, group discussions and public interpretation sessions with local actors.

IMPLEMENTATION & PRODUCTION

PROBLEMS

Technical tools need to be usable by non-experts
Open-source hardware and data workflows can be difficult for citizens, associations and local actors to use without clear documentation and support.

SOLUTIONS

Prototype adaptation and testing
Adapt and test the senseBox:bike setup for field deployment, including alternative mounting solutions for different bicycle configurations.

PROBLEMS

Data collection requires trust and continuity
Recruiting volunteers, installing sensors, supporting users and maintaining participation over time requires a clear and reliable process.

SOLUTIONS

Supported community data collection
Recruit volunteers, install sensors, provide guidance and maintain ongoing support so that participants can collect sensor data during everyday cycling trips.

PROBLEMS

Sensor data require calibration and validation
Measurements require further calibration, processing and validation before their accuracy and robustness can be fully established.

SOLUTIONS

Iterative calibration and processing
Document and refine sensor calibration, data-cleaning and event-detection procedures, including future validation against reference measurements or video.

PROBLEMS

Data only becomes useful after interpretation
Collecting data is not enough. The project needs structured analysis, collective interpretation and communication formats to turn measurements into action.

SOLUTIONS

Open documentation
Document the hardware, software, data workflow, workshop formats and communication outputs so the model can be analyzed, commented and reused in other cities.

ECOSYSTEM PARTNERS

PROBLEMS

Urban mobility challenges involve many actors
Cycling policies are shaped by municipalities, universities, cycling associations, local communities, makers and infrastructure actors.

SOLUTIONS

Ecosystem mapping
Map the local cycling landscape, identify key actors, contact partners and conduct snowball interviews to understand the local mobility ecosystem.

PROBLEMS

Collaboration needs clear roles
Without clear coordination, partner involvement can become fragmented, making it harder to move from data collection to recommendations and long-term action.

SOLUTIONS

Partner-based interpretation
Bring universities, cycling associations, citizens, local actors and makers into the interpretation process to validate findings and generate shared meaning.

PROBLEMS

Evidence must be connected to policy dialogue
Citizen-generated data has more impact when it is linked to public presentations, partner validation and policy recommendation processes.

SOLUTIONS

Policy recommendations
Transform evidence and emerging findings into recommendations that can support institutional dialogue and improvements for cycling.

COMMUNITY & CUSTOMERS

PROBLEMS

Communities need accessible entry points
Citizens may care about cycling conditions but need simple ways to participate, understand the data and contribute to advocacy.

SOLUTIONS

Volunteer participation model
Invite cyclists to contribute through everyday rides, sensor use, workshops and shared interpretation activities.

SOLUTIONS

Campaign materials for public awareness
Use posters, stickers, visuals and online dissemination to communicate key findings in accessible formats.

SOLUTIONS

Community advocacy toolkit
Provide reusable resources so local communities can replicate the process, communicate findings and advocate for safer cycling conditions.

SUSTAINABLE PLANS

SUSTAINABILITY PLAN

DATA-RIDE sensing and participation protocol
A tested campaign protocol combining local question-setting, sensor preparation and installation, translated app and instructions, volunteer data collection, data processing, collective interpretation, mapping and communication.

SUSTAINABILITY PLAN

Replicable DATA-RIDE toolkit
Maintain the toolkit as an open resource that can be adapted by other neighbourhoods, cities and cycling communities.

SUSTAINABILITY PLAN

Open-source maintenance
Keep hardware, software and documentation open, allowing future improvements, local adaptation and independent replication.

SUSTAINABILITY PLAN

Long-term partner network
Build continuity with universities, cycling associations, makers, public institutions and local actors to support future data rides and advocacy cycles.

SUSTAINABILITY PLAN

Community ownership
Support communities in using the data, visuals and recommendations beyond the project period, enabling continued advocacy and local action.

KNOWLEDGE

KNOWLEDGE OUT

Tested campaign protocol
Practical knowledge on participant onboarding, sensor installation, local adaptation (App translation and mounting system prototype), data collection, processing, collective interpretation and communication.

KNOWLEDGE IN

Technical and methodological knowledge
Open-source hardware, sensor calibration, data collection methods, mapping, workshop facilitation and visual communication.

KNOWLEDGE IN

Local cycling experience
Input from cyclists, citizens and associations about routes, unsafe areas, visibility issues, overtaking and infrastructure gaps.

KNOWLEDGE IN

Institutional and policy context
Knowledge from local actors, universities and public institutions about planning processes, mobility priorities and possible interventions.

KNOWLEDGE OUT

Shared insights and emerging hypotheses
Collective interpretations developed through analysis, public presentations and group discussions.

KNOWLEDGE OUT

Communication and campaign materials
Posters, stickers, visuals, slogans and online content that translate data into public awareness.

KNOWLEDGE OUT

Citizen-generated dataset
A dataset based on everyday cycling journeys, documenting conditions such as safety, interactions, visibility and mobility experience

CC BY 4.0

Co-funded by the European Union,
2024-2026, GA 101135986

PROJECT INFORMATION

Mobility

This blueprint is used to document the problems addressed and the solutions developed within a project. It does not belong to the factory itself, but rather to a specific product created by the factory, encompassing everything related to that product or service.

Project Description

DATA-RIDE is a policy-driven citizen science project based on the open-source senseBox:bike platform, combining multiple data sources to support evidence-based advocacy and policy recommendations for safer urban cycling.

Vision

To make everyday cycling conditions visible, understandable and actionable by turning citizen-generated mobility data into evidence for safer, more inclusive and policy-driven urban cycling.

Roles & Competences

- Artur Kim Shum — re:edu; design, coordination, data analysis, visualization and communication.
- Laura Sobral — project coordination, advocacy, communication and community engagement.
- Laurent Dupont — Université de Lorraine / ERPI / Lorraine Fab Living Lab; local partnership and academic guidance.
- Anaëlle Hily — Université de Lorraine; local support, coordination and implementation in Nancy.

re:edu

- Felix Erdmann — technical development and implementation.
- Paula Scharf — technical development and implementation.

Global Innovation Gathering — GIG

- Ricardo Ruiz — dissemination, knowledge transfer and international network.
- Melanie Chan — dissemination, knowledge transfer and international network.

Organizations involved

- re:edu
- Global Innovation Gathering — GIG
- Université de Lorraine
- Lorraine Fab Living Lab
- LAUDS Factories
- senseBox
- local cycling associations and actors in Nancy
- citizens and volunteer cyclists who collected data

Information on Blueprint Canvas

This Blueprint Canvas has been developed within the WP5 LAUDS Factories.

CC BY 4.0 International. Bassi, E., Subet, M., Cangiano, S., Veschetti, D. (2026). LAUDS Factories Blueprint V3.

KEY LEARNINGS

KL01

Collective interpretation creates impact
Workshops, public presentations and group discussions help transform raw data into shared understanding and useful advocacy messages.

KL02

Data collected through everyday rides shows cycling conditions that are often invisible in traditional planning processes because it's based on the real experiences

KL03

Cycling safety depends on interactions, not infrastructure alone
Safety emerges from the relationship between cyclists, drivers, pedestrians, street design, parking conditions and traffic behaviour.

MAIN ISSUES

MI01

Data calibration and interpretation require more preparation time
Sensor calibration, data cleaning and collective interpretation require dedicated preparation time before findings can be translated into reliable insights.

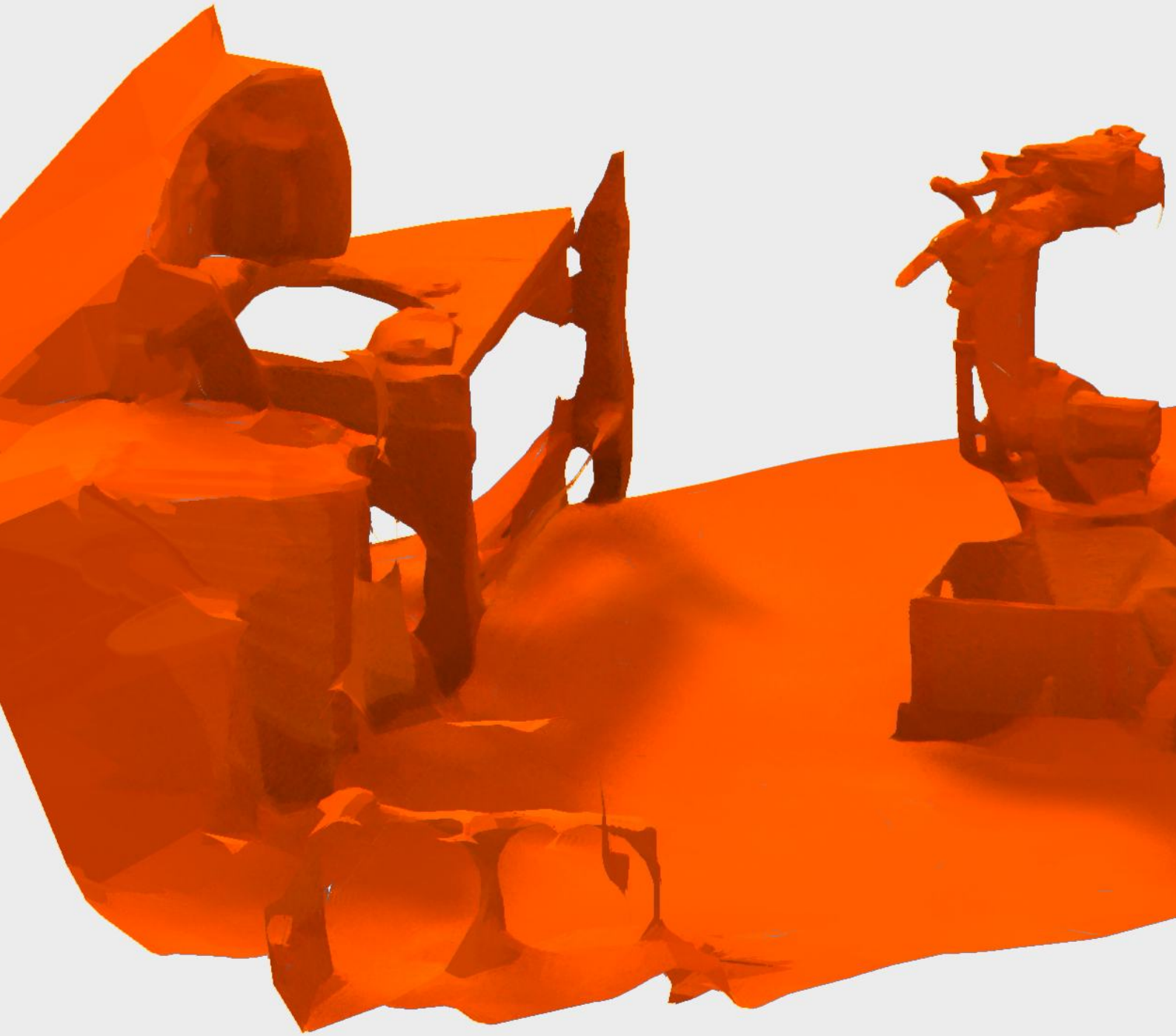
MI02

Bike compatibility requires adaptable mounting solutions
Different bicycle models require different support designs, mounting strategies and testing procedures to ensure the sensor can be safely and reliably installed.

CC BY 4.0

LAUDS Factories

[DATA-RIDE] Blueprint



Laura Sobral, Artur Kim Shum, Laurent Dupont, Anaëlle Hily